CVS // INDUSTRIAL SYSTEMS FIELD GUIDE

INTERACTIVE PORTFOLIO // DOCUMENT EDITION

CHANDER VINAYAK SHARMA

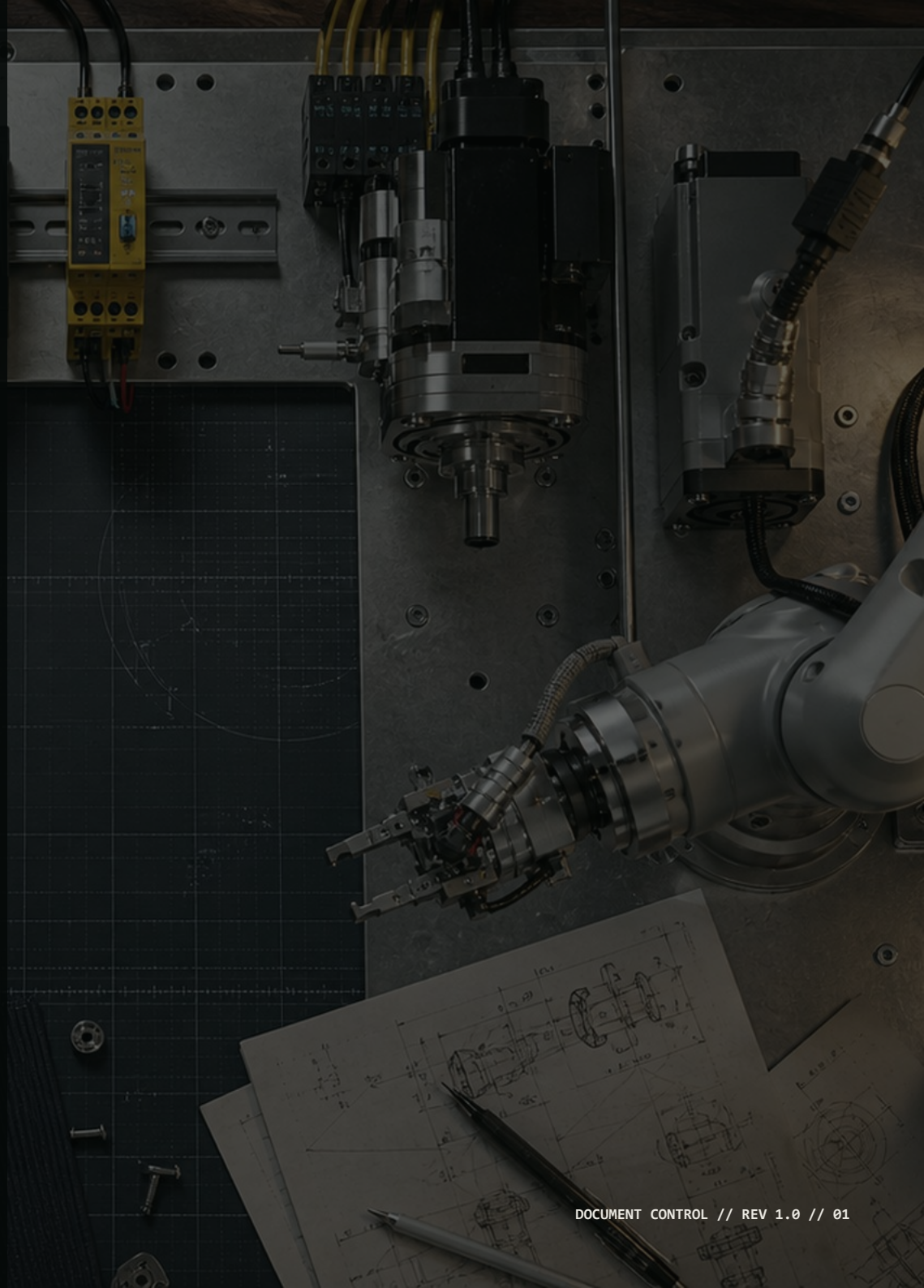
INDUSTRIAL AUTOMATION ENGINEER // ROBOTICS

Industrial automation and robotics systems builder translating requirements into deterministic control, virtual commissioning evidence, industrial data products and explainable engineering workflows.

OPEN LIVE SYSTEMS WORLD →

SCAN // LIVE 3D EXPERIENCE

06	02	01
ENGINEERING SYSTEMS	DEPLOYED VENTURES	ACTIVE RESEARCH



DOCUMENT CONTROL // REV 1.0 // 01

DOCUMENT MAP

A portfolio built as an engineering argument.

Each system is framed by the operating problem, the control or software method, the evidence retained and the physical boundary that still remains.

SYS.01 // MOTION / SAFETY

Industrial Motion and Safety Bench

10 ms // PLC TASK

SYS.02 // SCADA / DIGITAL TWIN

Material Handling Micro-Cell

168 / 0 // PLC VALIDATION

SYS.03 // REAL-TIME / CIA 402

Embedded Real-Time Motion Controller

1 ms // CYCLIC MODEL

SYS.04 // RELIABILITY / DIAGNOSTICS

Reliability and Predictive-Maintenance Layer

97.5% // WINDOW ACCURACY

SYS.05 // AMR / FLEET

AMR and Fleet Integration Stack

100 / 100 // DOCKING MISSIONS

SYS.06 // EVENT LOG / STORAGE

KairoLog

GO // BROKER CORE

03

PROFILE + EXPERIENCE

Controls, robotics, data and operating discipline

11

VENTURES

BuildTrace Beta + LaTrack // v0.1 deployed

12

RESEARCH

ML in autonomous navigation // bibliometric mapping

13

PODCAST

Industry Signal // company and engineering conversations

OPEN LIVE EXPERIENCE →

OPERATOR PROFILE

Systems builder across controls, robotics and industrial data.

A systems-focused engineering profile spanning industrial automation, robotics, real-time software, reliability and connected industrial data.

PLC & INDUSTRIAL CONTROL

TwinCAT 3, Beckhoff, Structured Text, IEC 61131-3, PLCopen-style motion, state machines, HMI and alarm handling.

ROBOTICS & MOTION

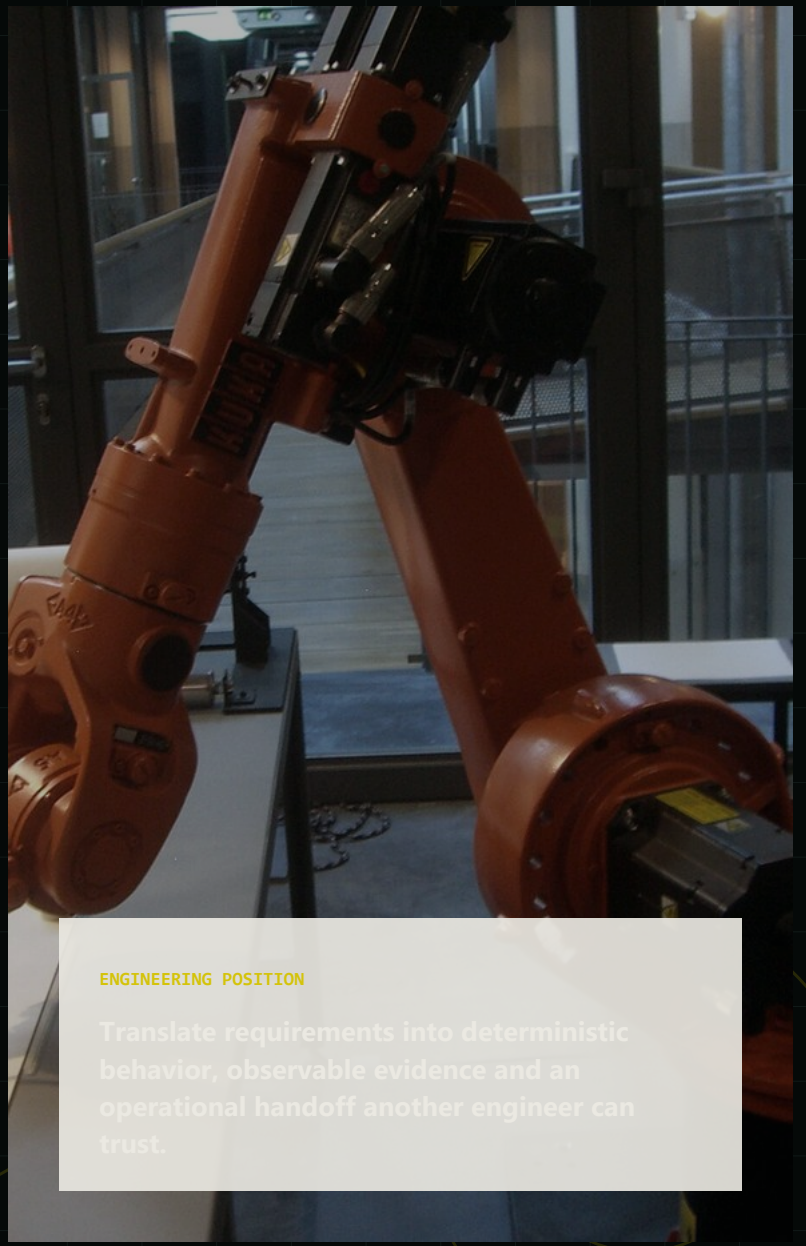
ROS 2, Nav2, Gazebo, C++17, CiA 402, EtherCAT, CANopen, deterministic cyclic control and recovery behavior.

INDUSTRIAL DATA & SOFTWARE

SCADA/HMI, OPC UA, MQTT, InfluxDB, Grafana, Python, NumPy, Pandas, reliability analytics, SQLite, Go and HTTP APIs.

ENGINEERING PRACTICE

Virtual commissioning, digital twins, FAT/SAT evidence, trace logging, fault injection, GitHub Actions, CMake, FMEA/FTA and technical documentation.



ENGINEERING POSITION

Translate requirements into deterministic behavior, observable evidence and an operational handoff another engineer can trust.

OPERATING RECORD

Experience grounded in operations and machine systems.

The common thread is disciplined troubleshooting: establish the intended process, observe the real behavior, preserve evidence and communicate the next action clearly.

2023-2026 // MISSISSAUGA, CANADA

Technical Operations Analyst

Walmart Canada

- Investigated operational issues across repeatable workflows, separating symptoms from root causes and escalating defects with clear reproduction details.
- Performed data-quality checks and reconciliations so teams could make decisions from consistent operational information.
- Maintained process documentation and translated changing requirements into practical steps for cross-functional teams.
- Built a disciplined understanding of reliability, service ownership, prioritization and communication in a high-volume operating environment.

ENGINEERING REPORT // FULLMATIK SRO

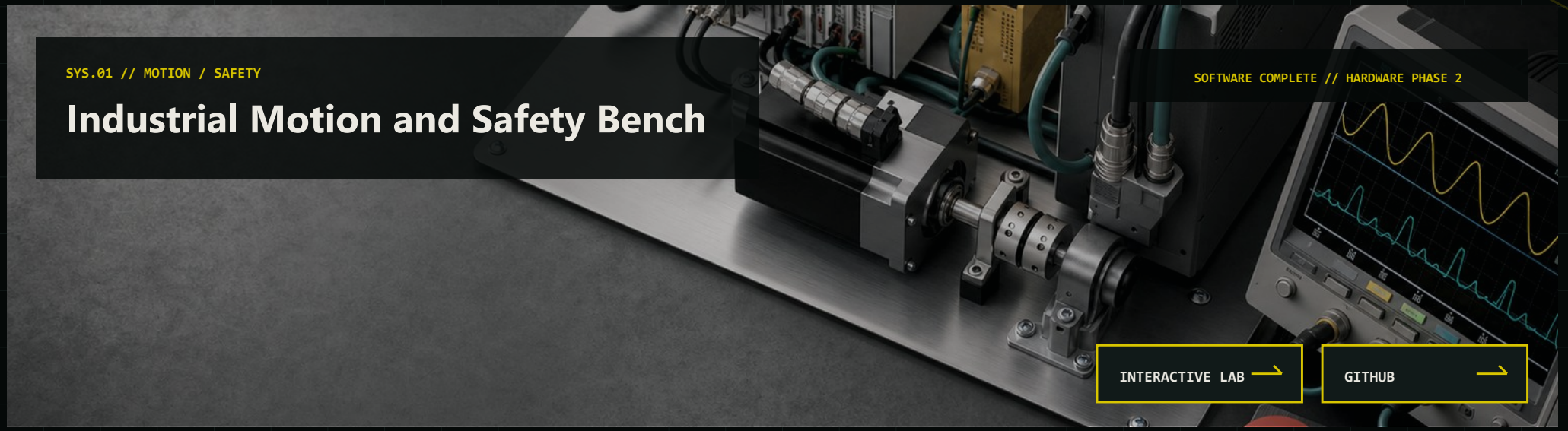
Bottle Unscrambler Analysis

Machine systems study

- Mapped the bottle flow from infeed to orientation and discharge, identifying the stages that determine throughput and repeatability.
- Reviewed mechanical components, sensing points and control interactions that influence jams, misfeeds and changeover behavior.
- Connected machine function to practical automation, reliability and production requirements in a structured technical report.

REQUIREMENTS — CONTROL MODEL — FAULT HANDLING — EVIDENCE — HANDOFF

A repeatable engineering workflow applied across automation, robotics, reliability and connected industrial software.



SYS.01 // MOTION / SAFETY

Industrial Motion and Safety Bench

SOFTWARE COMPLETE // HARDWARE PHASE 2

INTERACTIVE LAB →

GITHUB →

Industrial Motion and Safety Bench

TwinCAT 3 virtual commissioning, PLC motion logic, safety interlocks and FAT evidence.

10 ms	16 / 16	25,445
PLC TASK	SOFTWARE FAT	SCOPE SAMPLES

WHY

De-risk a new servo axis and its commissioning sequence before the DIN-rail cabinet, drive and safety chain are purchased.

HOW

Translate axis requirements into modular IEC 61131-3 control, explicit operating modes, guarded commands, homing and recovery logic, then exercise the full sequence in a repeatable software FAT.

RESULT

A reviewable baseline that exposes normal motion, rejected commands, alarms and recovery behavior while hardware choices can still be changed inexpensively.

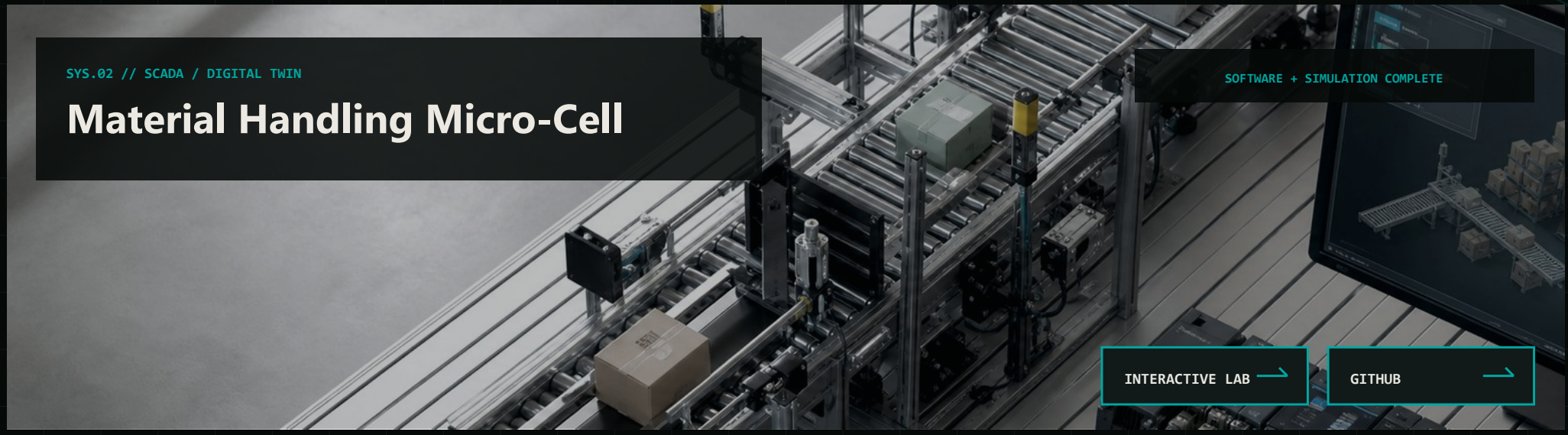
CONTROL / DATA ARCHITECTURE

```
ConfigPackage -> SafetyManager -> AlarmManager -> ModeManager -> CommandParser -> HomingManager -> AxisManager ->
TraceLogger -> HMIModel -> TestHarness.
```

EVIDENCE // ADS port 852; 19.772 s retained FAT run; CSV, JSON, PNG, XLSX and SHA-256 evidence package.

TRUTHFUL HARDWARE BOUNDARY

Safety-oriented controls demonstration only. Hardware SAT, servo commissioning, stopping-time measurement and certified safety validation remain Phase 2.



SYS.02 // SCADA / DIGITAL TWIN

Material Handling Micro-Cell

SOFTWARE + SIMULATION COMPLETE

INTERACTIVE LAB →

GITHUB →

Material Handling Micro-Cell

PLC control, semantic OPC UA, historian/KPI service, HMI/SCADA and a visual sorting-cell twin.

168 / 0	10 / 0	0 / 0
PLC VALIDATION	TEST HARNESS	BUILD ERR / WARN

WHY

Evaluate a sorting-cell control concept and its OT/IT data contract before conveyor hardware is commissioned.

HOW

Model scan, routing, diverter, jam and alarm behavior as modular PLC functions while keeping HMI, OPC UA, historian and Grafana views aligned to one semantic tag model.

RESULT

A complete software demonstration in which faults and manual recovery are replayable, telemetry is inspectable and commissioning assumptions are visible before procurement.

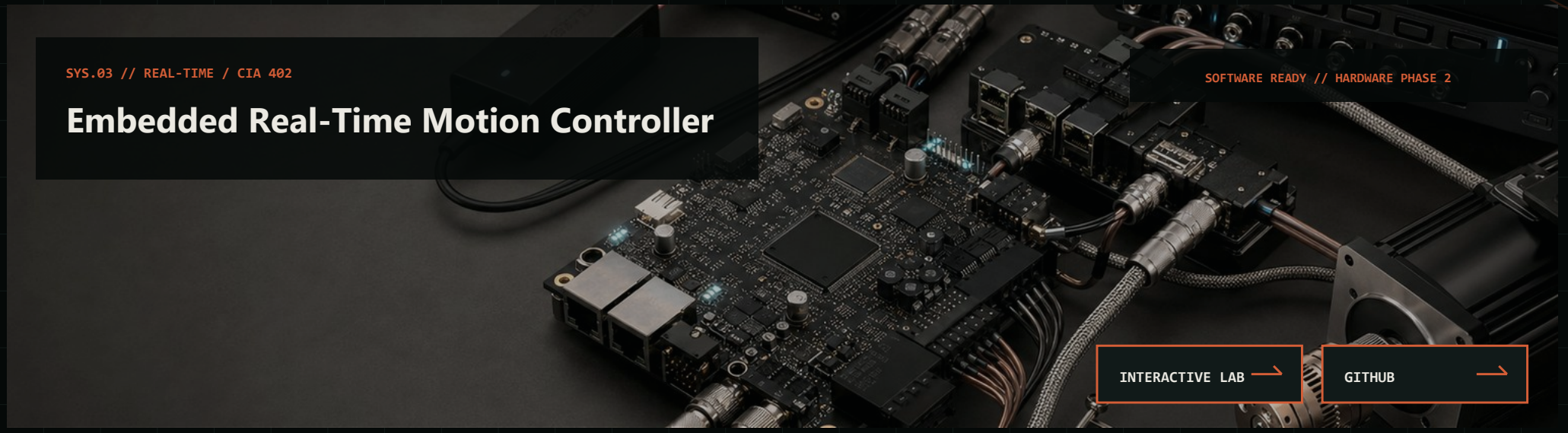
CONTROL / DATA ARCHITECTURE

`FB_LineSupervisor -> FB_StationController -> FB_RoutingLogic -> FB_DiverterController -> FB_JamDetector -> FB_AlarmManager -> FB_KPIService -> Historian.`

EVIDENCE // HMI prototype, visual digital twin and historian preview endpoints pass the software demo smok

TRUTHFUL HARDWARE BOUNDARY

Physical conveyor commissioning, certified SIL/PL/CE validation and signed production FAT/SAT remain Phase 2.



SYS.03 // REAL-TIME / CIA 402

Embedded Real-Time Motion Controller

SOFTWARE READY // HARDWARE PHASE 2

INTERACTIVE LAB →

GITHUB →

Embedded Real-Time Motion Controller

Deterministic C++ motion core with CiA 402 behavior, watchdogs, fault injection and timing evidence.

1 ms	21 / 21	45
CYCLIC MODEL	SCENARIOS	C++ CHECKS

WHY

Exercise embedded motion-control failure paths and timing instrumentation before EtherCAT or CANopen hardware integration.

HOW

Build a portable C++17 cyclic controller around guarded CiA 402 transitions, deterministic scenarios, watchdogs, fault injection, trace capture and replayable timing reports.

RESULT

A software release gate with quantified missed-cycle detection, deadline-overrun behavior, recovery latency and two-axis motion acceptance.

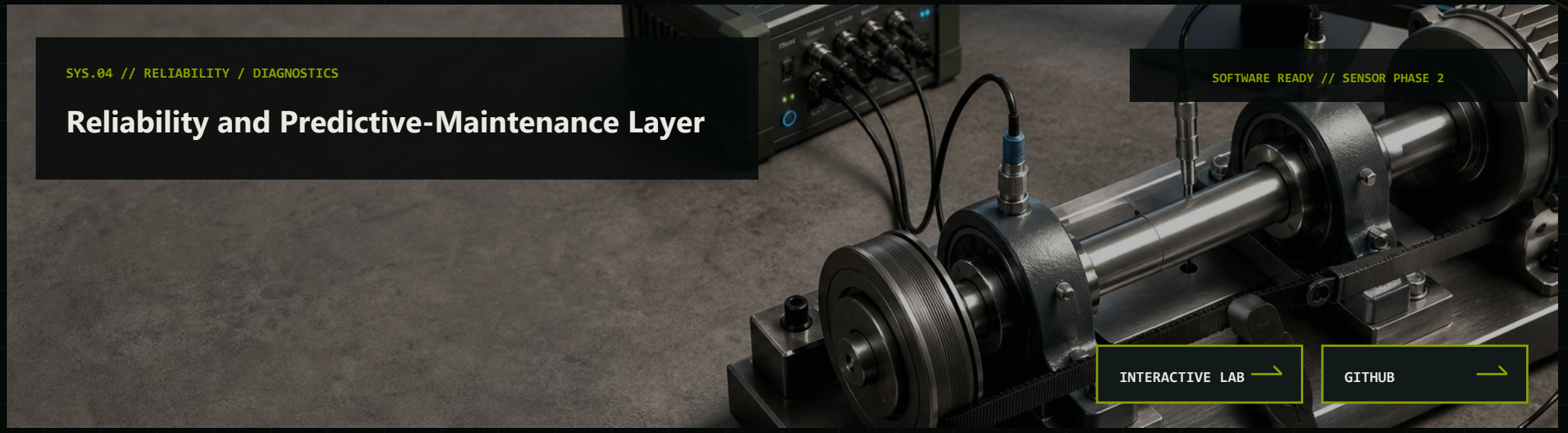
CONTROL / DATA ARCHITECTURE

Scheduler -> Axis Core -> CiA 402 State Machine -> Bus Manager -> Watchdog -> Fault Injector -> Telemetry / Trace Store.

EVIDENCE // Two injected missed cycles and two deadline overruns; zero-cycle detection latency, seven cycl

TRUTHFUL HARDWARE BOUNDARY

Physical bus commissioning, real encoder evidence, hardware stop-path verification and target real-time measurements remain Phase 2.



SYS.04 // RELIABILITY / DIAGNOSTICS

Reliability and Predictive-Maintenance Layer

SOFTWARE READY // SENSOR PHASE 2

INTERACTIVE LAB →

GITHUB →

Reliability and Predictive-Maintenance Layer

Explainable condition monitoring, seeded faults, maintenance actions and connected telemetry.

97.5%	5 / 5	144,000
WINDOW ACCURACY	FAULT CLASSES	SAMPLES

WHY

Move from raw simulated telemetry to a defensible maintenance action without presenting synthetic sensors as field measurements.

HOW

Quality-gate vibration, current and temperature windows, compute classical features, compare them with a robust median/IQR baseline and map explainable rules to alarms, spares and work orders.

RESULT

Five seeded fault classes are diagnosed with quantified recall and false-alert behavior, while every result carries a clear physical-sensing boundary.

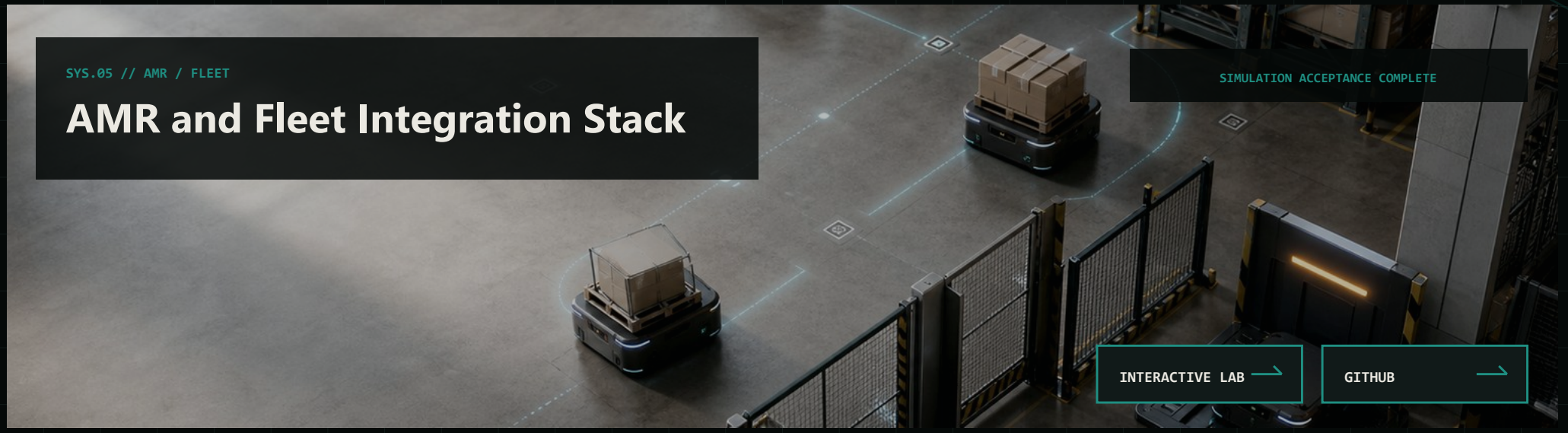
CONTROL / DATA ARCHITECTURE

Telemetry Gate -> Feature Extraction -> Median/IQR Baseline -> Diagnostic Rules -> Alarm Rationalization -> Maintenance Matrix -> Store-and-Forward.

EVIDENCE // Nineteen Python tests pass across 240 five-second diagnostic windows, with 95.65% seeded-fault

TRUTHFUL HARDWARE BOUNDARY

Physical accelerometer commissioning, measured baselines, field predictive accuracy and production FAT/SAT remain Phase 2.



AMR and Fleet Integration Stack

Mission lifecycle, docking, mutual exclusion, degraded modes and operations telemetry.

100 / 100	0	5 / 5
DOCKING MISSIONS	EXCLUSION VIOLATIONS	DEGRADED MODES

WHY

Test fulfillment-style AMR missions, station handoffs and recovery behavior before physical robots and safety chains are commissioned.

HOW

Coordinate mission intake, station locks, navigation lifecycle, docking, degraded modes and telemetry through one observable fleet state model.

RESULT

A repeatable non-hardware acceptance package covering mission completion, mutual exclusion, docking consistency and controlled degraded operation.

CONTROL / DATA ARCHITECTURE

```
Mission API -> Fleet Orchestrator -> Station Locks -> Nav2 Lifecycle -> Docking State Machine -> Telemetry Bridge -> Operations Dashboard.
```

EVIDENCE // Software acceptance covers mission flow, station handoff, recovery behavior and non-hardware p

TRUTHFUL HARDWARE BOUNDARY

Physical AMR commissioning, certified safety-chain validation, physical video and simulation-vs-real comparison remain open.



SYS.06 // EVENT LOG / STORAGE

KairoLog

CORE BROKER FOUNDATION COMPLETE

INTERACTIVE LAB →

GITHUB →

KairoLog

Kafka-inspired Go commit log with partitions, indexes, offsets, groups and durable state.

GO	13	JSONL
BROKER CORE	TEST FILES	DURABLE STATE

WHY

Provide a durable event and telemetry backbone for automation services before a larger distributed-broker rollout.

HOW

Implement append-only segmented storage, byte-position indexes, partitions, consumer offsets, group assignment, heartbeats and durable recovery directly in Go.

RESULT

A transparent single-node broker foundation where ordering, replay and coordination can be traced from HTTP ingress to recovered consumer state.

CONTROL / DATA ARCHITECTURE

HTTP Ingress -> Topics -> Partitions -> Append-only Segments + Byte Indexes -> Consumer Offsets -> Group Assignment -> Heartbeats.

EVIDENCE // File-backed offsets, assignments and group registry support deterministic rebalance, stale-mem

TRUTHFUL HARDWARE BOUNDARY

The current boundary is a strong single-node broker and storage foundation. Replication, leader election and multi-broker operations remain roadmap work.

DEPLOYED BETA SYSTEMS

Two live products. One market-learning loop.

BuildTrace Beta and LaTrack are deployed software products. The current operating loop is client outreach, market validation and finding repeatable product-market fit.

VNT.01 // V0.1 DEPLOYED

BuildTrace Beta

A multilingual machine-lifecycle workspace for SME machine builders, automation integrators and OEMs. It turns delivered equipment into a secure digital record covering engineering documents, controlled customer access, service history, software versions, spare parts and quote workflows.

TECHNICAL SPINE

Next.js + TypeScript // NestJS + Fastify // PostgreSQL + Prisma // Supabase

CURRENT // MARKET VALIDATION, CLIENT OUTREACH AND REPEATABLE PRODUCT-MARKET FIT.

VNT.02 // V0.1 DEPLOYED

LaTrack

A role-aware university laboratory access system that helps students find and request the right lab while giving faculty a time-stamped record of approvals, scheduled windows, check-in, occupancy and usage.

TECHNICAL SPINE

Role-aware requests // faculty approval // access windows // audit records

CURRENT // INSTITUTIONAL OUTREACH, WORKFLOW VALIDATION AND REPEATABLE PRODUCT-MARKET FIT.

DEPLOY -> OBSERVE -> INTERVIEW -> PRIORITIZE -> REPEAT

Both v0.1 products are deployed. Current work is market validation, client outreach and finding repeatable product-market fit.

DISCUSS A VENTURE →

RESEARCH CHANNEL // IN PROGRESS

Quantitative Mapping of Machine Learning in Autonomous Navigation

An in-progress bibliometric study that treats academic literature as a measurable system: cleaning publication metadata, mapping collaboration and co-citation networks, and tracking emerging technical themes.

DATA PIPELINE

Publication records are normalized, deduplicated and enriched so authors, institutions, countries, citations, keywords and publication years can be compared reproducibly.

NETWORK MODEL

Co-authorship, institutional collaboration, co-citation and keyword co-occurrence graphs expose communities, influential bridges and the structure of knowledge production.

TEMPORAL SIGNAL

Growth curves and topic evolution reveal when methods move from emerging signals to established research directions, supporting a manuscript intended for peer review.



INDUSTRY SIGNAL // EPISODE 01

Technical conversations with the people moving industry forward.

I host and invite directors, CEOs, founders, plant leaders, product heads and engineers to explain the companies, products, systems and decisions behind modern industry. The format gives companies a thoughtful stage for their story while giving aspiring engineers a grounded view of where the industry is moving.





COMPANY STORY

History, vision and the problem the company exists to solve.

PRODUCTION SYSTEM

Product catalogues, lines, automation choices and Industry 4.0 decisions.

ENGINEERING TRANSFER

Recent milestones, lessons and practical direction for aspiring engineers.

INVITE / COLLABORATE →

EXTERNAL I/O

Build systems that behave clearly and prove what they do.

Available for robotics, automation, controls, motion, industrial data and systems engineering roles, as well as serious technical collaboration.

EMAIL	chander11132@gmail.com
GITHUB	github.com/chad416
LIVE PORTFOLIO	chad416.github.io/chander-portfolio

EMAIL CHANDER →

OPEN LIVE PORTFOLIO →

SYSTEMS INDEX // LIVE ROUTES

- SYS.01 Industrial Motion and Safety Bench
- SYS.02 Material Handling Micro-Cell
- SYS.03 Embedded Real-Time Motion Controller
- SYS.04 Reliability and Predictive-Maintenance Layer
- SYS.05 AMR and Fleet Integration Stack
- SYS.06 KairoLog

SCAN // ENTER SYSTEMS WORLD

DETERMINISTIC CONTROL // OBSERVABLE EVIDENCE // OPERATIONAL THINKING